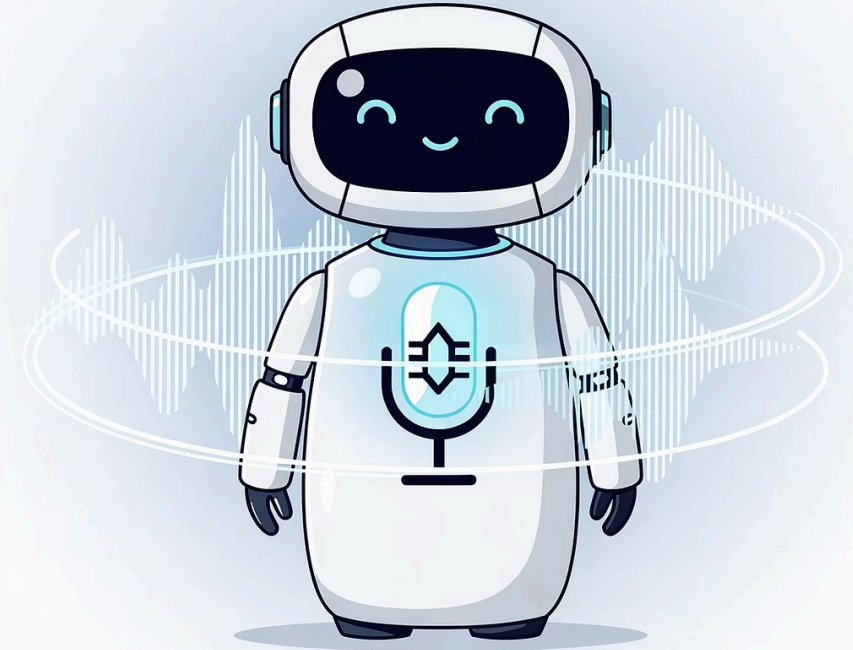


Session 16:

AI That Hears & Speaks - Voice AI

Today, you'll discover how AI listens and speaks. We'll explore how sound waves become words, how AI figures out meaning, and how it generates lifelike voices. You'll get hands-on with ElevenLabs to create your own synthetic speech.



Objectives

01

Define

Voice AI and explain why hearing and speaking are hard for computers.

02

Describe

The Voice AI framework: **Hear** → **Process** → **Speak**.

03

Explain

What happens inside each stage: sound waves, microphone, ASR/STT, acoustic features, phonemes, NLU, response, TTS, tokens, prosody, waveform.

04

Identify

Common Voice AI tasks (speech-to-text, text-to-speech, voice commands, speaker recognition).

05

Recognize

Challenges in Voice AI (noise, accents, homophones, bias, privacy).

06

List

Real-world Voice AI applications (voice assistants, audiobooks, accessibility, translation).

07

Use

ElevenLabs to generate synthetic speech and experiment with voice settings.

08

Add

A Voice AI-focused prompt to your Prompt Library.

Recap: What We Learned in Session 15

Before we dive in, let's quickly revisit what we covered last time.

Key Idea	What It Means
Computer Vision (CV)	AI that interprets images and videos.
See → Process → Act	The three-stage framework for visual AI.
Pixels, Features, Patterns, Prediction	How AI turns images into understanding.
Key insight	CV is pattern matching, not true seeing.

 **Today we move from seeing to hearing and speaking - Voice AI.**

The Big Question

Humans filter noise, understand context, and adapt to accents naturally. Computers hear only numbers - sound waves turned into data. How can a machine figure out what's being said and respond?

How does AI turn Voice into meaningful words and then back into spoken language?



What is Voice AI?

Voice AI is the branch of AI that gives computers the ability to hear, interpret, and generate spoken language. It combines **two major technologies**:

ASR / STT

Automatic Speech
Recognition / Speech-to-Text
- listening and converting
speech to text.

TTS

Text-to-Speech - turning text
into spoken audio.

Think of Voice AI as a translator between sound and text. A microphone turns invisible sound waves into a digital squiggle called a waveform. Voice AI decodes that waveform into words, figures out what you want, and can speak back to you in a synthetic voice.

Where You See Voice AI Every Day



Voice Assistants

Siri, Alexa, Google Assistant



Voice Typing

Voice typing on your phone



Audiobooks

Audiobooks narrated by AI voices



Live Captions

Live captions on videos and video calls



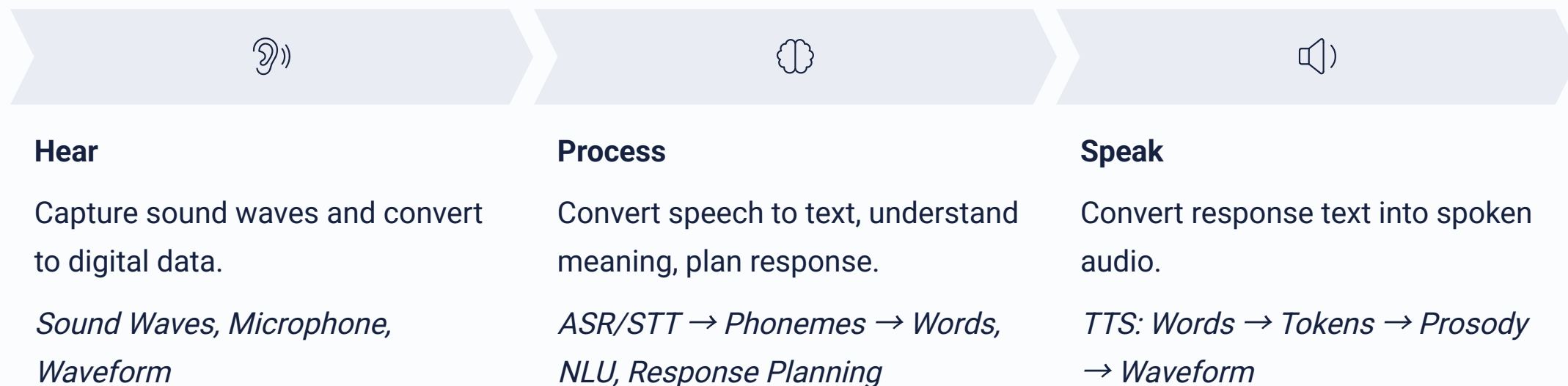
Translation Apps

Translation apps that speak aloud

📌 **Key insight:** Voice AI does not "hear" or "understand" like you do. It finds mathematical patterns in sound waves and matches them to words it has seen millions of times. It's pattern matching, not true comprehension.

How Voice AI Works: Hear → Process → Speak

Voice AI follows a three-stage framework that matches what you learned in Session 1 (Sense → Think → Act).



Inside the "Hear" Stage

Term	What it means	Simple analogy
Sound Wave	Vibrations in the air that your ear detects.	A wiggly line that represents your voice.
Microphone	A sensor that converts sound waves into an electrical signal.	Like your ear turning vibrations into signals for your brain.
Waveform	A visual graph of sound over time – the "shape" of the sound.	A picture of wiggles; different words have different wiggle shapes.

📄 **Think of it like a flipbook:** The faster you flip pages, the smoother the motion appears. Voice AI samples sound thousands of times per second to capture all the details of speech. This is called the **sampling rate** - how many measurements the computer takes each second. More samples = better quality.

Inside the "Process" Stage

This is where the real work happens. The "Process" stage has three parts: **ASR/STT**, **NLU**, and **Response Planning**.

01

ASR/STT

Automatic Speech Recognition / Speech-to-Text - converts the waveform into written words.

02

NLU

Natural Language Understanding - figures out what you want: intent, entities, sentiment.

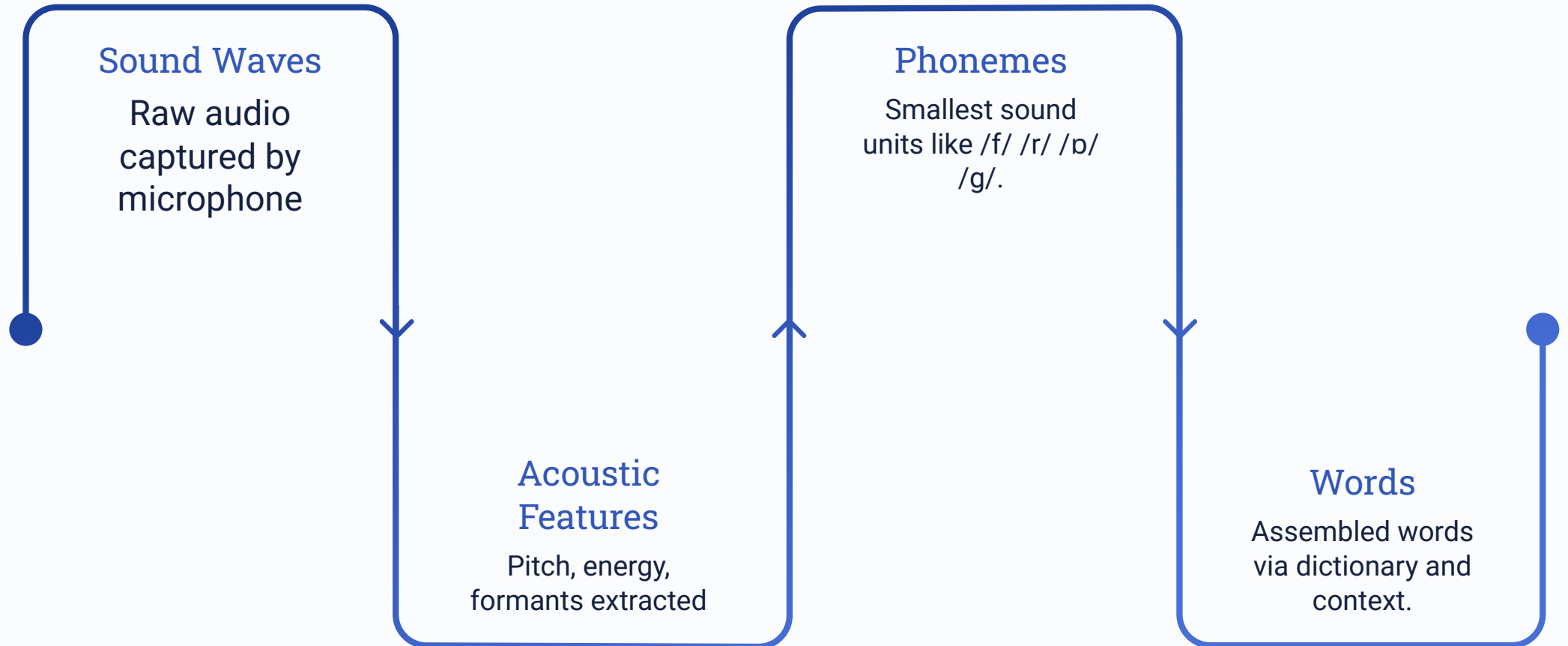
03

Response Planning

The AI decides what to say or do next: fetch information, trigger an action, or formulate a text response.

Part 1: ASR/STT - Sound to Text

ASR/STT converts the waveform into written words using this four-step flow:



Step	What happens	Example
Sound Waves	Raw audio captured by microphone.	The wiggly waveform of "frog".
Acoustic Features	AI extracts clues from the sound: pitch, energy, frequency patterns.	Like finding edges and colors in an image (Session 15).
Phonemes	The smallest units of sound in a language. English has about 40 phonemes.	/f/ + /r/ + /ɒ/ + /g/ for "frog".
Words	AI assembles phonemes into complete words using a dictionary and context.	"frog"

Part 2: NLU - Natural Language Understanding

Once ASR/STT produces text, NLU figures out what you want



Intent

What action do you want? (e.g., set a reminder, get weather, play music)



Entities

Important details (e.g., task = call home, time = 6 PM)



Sentiment

Positive, negative, or neutral tone

Part 3: Response Planning

The AI decides what to say or do next. It might fetch information (weather data), trigger an action (turn on lights), or formulate a text response.

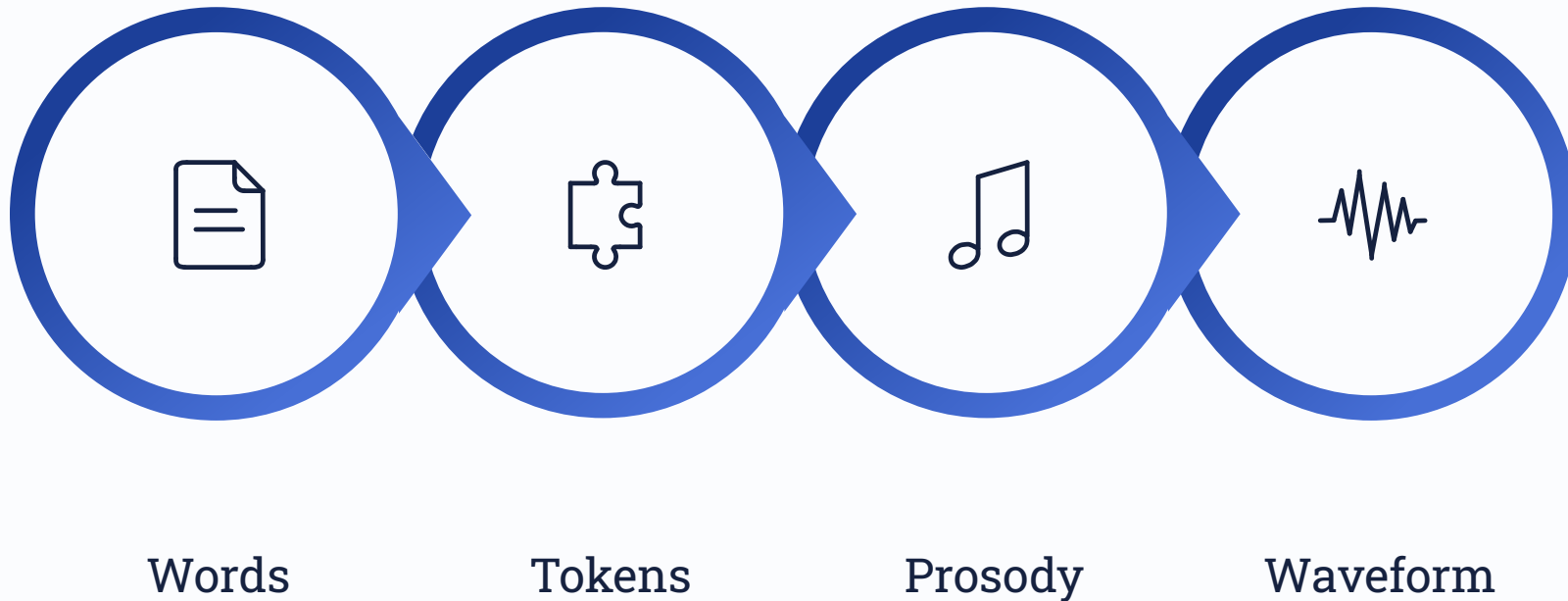
Example: "Remind me to call home at 6 PM"

Step	What happens
Sound Waves	Microphone records your voice.
Acoustic Features	AI extracts pitch and frequency patterns.
Phonemes	/r/ /ɪ/ /'m/ /aɪ/ /n/ /d/ /m/ /i:/ /t/ /ə/ /k/ /ɔ:/ /l/ /h/ /əʊ/ /m/ /æ/ /t/ /s/ /ɪ/ /k/ /s/ /p/ /i:/ /'ε/ /m/
Words	"Remind me to call home at 6 PM"
NLU	Intent = set_reminder, Task = call home, Time = 6 PM
Response Planning	Text response: "Okay, I'll remind you to call home at 6 PM."

 **Important:** The AI doesn't "know" what a reminder is or what "home" means. It has learned that the words "remind me to [task] at [time]" usually mean the user wants a notification scheduled.

Inside the "Speak" Stage

Once the AI has a text response, it uses **Text-to-Speech (TTS)** to turn words back into sound using this four-step flow:



Step	What happens	Analogy
Words	The text response from the Process stage.	"Timer set for 5 minutes."
Tokens	Text is broken into small pieces (like Session 8 and 14).	["Timer", "set", "for", "5", "minutes"]
Prosody	The "music" of speech – rhythm, pitch, stress, and emotion. AI predicts how the words should sound.	Like a director telling an actor: "Say this calmly and clearly."
Waveform	AI generates the actual sound wave and plays it through the speaker.	The voice you hear.

- ❏ **Prosody is what makes AI voices sound human.** Without it, speech would be flat and robotic. Modern TTS (like ElevenLabs) uses deep learning to predict natural prosody - where to pause, which words to emphasize, and what tone to use.

See It All Together: "Hey Siri, what's the weather?"

Stage	What happens
Hear	Microphone captures "Hey Siri, what's the weather?" as a waveform.
Process (ASR/STT)	Sound Waves → Acoustic Features → Phonemes → Words: "Hey Siri what's the weather"
Process (NLU)	Intent = get_weather; Location = current location.
Process (Response)	AI fetches weather data, forms text: "It's currently 35 degrees and sunny."
Speak	Words → Tokens → Prosody → Waveform: AI speaks the response.

All of this happens in less than a second!

Mini Activity 1: Match the Voice AI Concept

Instructions: For each description, write which Voice AI concept it matches. Choose from: **Waveform**, **Acoustic Features**, **Phoneme**, **ASR/STT**, **TTS**, **NLU**, **Prosody**, **Tokens**.

#	Description	Concept
1	The process of converting spoken audio into written text.	_____
2	A visual graph of sound over time - the "shape" of the sound wave.	_____
3	The smallest unit of sound in a language (e.g., /b/ in "bat").	_____
4	The process of converting text into spoken audio.	_____
5	Clues extracted from sound: pitch, energy, frequency patterns.	_____
6	Figuring out the intent and meaning behind the words.	_____
7	The rhythm, pitch, and stress that make speech sound human.	_____
8	Small pieces that words are broken into for processing.	_____

Mini Activity 1: Answer Key

1 → ASR/STT

Automatic Speech
Recognition / Speech-
to-Text

2 → Waveform

Visual graph of sound
over time

3 → Phoneme

Smallest unit of sound

4 → TTS

Text-to-Speech

5 → Acoustic Features

Pitch, energy, frequency
patterns

6 → NLU

Natural Language
Understanding

7 → Prosody

Rhythm, pitch, and
stress

8 → Tokens

Small pieces words are
broken into

📌 These concepts work together in sequence: Hear (Waveform) → Process (ASR/STT: Acoustic Features → Phonemes → Words; NLU) → Speak (TTS: Words → Tokens → Prosody → Waveform).

Common Voice AI Tasks

Task	What it does	Framework example
Speech-to-Text (STT) / ASR	Converts spoken words into written text.	Hear: waveform of "Hello" → Process (ASR/STT): "Hello" text.
Text-to-Speech (TTS)	Converts written text into spoken audio.	Process: analyze text → Speak: generate synthetic voice.
Voice Commands	Understands and executes spoken instructions.	Hear: "Turn on lights" → Process: intent = control_device → Action.
Speaker Recognition	Identifies who is speaking based on voice characteristics.	Hear: voice sample → Process: match to stored voiceprint.
Voice Cloning	Creates a synthetic copy of a specific person's voice.	Hear: sample of person → Process: learn voice characteristics → Speak: new speech in that voice.

Challenges in Voice AI

Background Noise

Other sounds mix with the voice, confusing the AI.

A voice assistant in a car with music playing mishears commands.

Accents & Dialects

Pronunciation varies by region; AI may fail on unfamiliar accents.

AI trained on American English struggles with a Scottish accent.

Homophones

Words that sound the same but have different meanings.

"Write" vs. "right" - AI must use context to choose.

Bias in Training Data

If data lacks diverse voices, AI performs worse for some groups.

ASR/STT systems have higher error rates for women and people of color.

Privacy Concerns

Voice assistants are always listening; recordings may be stored.

Users worry about private conversations being recorded.

Emotional Tone

Sarcasm, anger, or excitement change how words are said.

"That's great" said flatly vs. enthusiastically – AI may miss the true meaning.

📌 **Key takeaway:** Voice AI is powerful but not perfect. Noise, accents, bias, and privacy are active areas of research and improvement.

Mini Activity 2: Spot the Voice AI Challenge

Instructions: For each scenario, identify which Voice AI challenge is present. Choose from: **Background Noise, Accent, Homophone, Bias, Privacy, Emotional Tone.**

#	Scenario	Challenge
1	A smart speaker in a busy kitchen hears "Add milk" but writes "Add silk."	_____
2	A voice assistant fails to understand someone from rural Ireland but works fine for a California accent.	_____
3	A user says "I need to book a flight" and the AI hears "I need to book a fight."	_____
4	An ASR/STT system has a 20% higher error rate for female voices than male voices.	_____
5	A user says "Oh, wonderful" sarcastically, but the AI thinks it's positive.	_____
6	A family discovers their smart speaker recorded a private conversation.	_____

Mini Activity 2: Answer Key

1 → Background Noise

Kitchen sounds confused "milk" with "silk."

2 → Accent

Rural Irish accent not in training data.

3 → Homophone

"flight" and "fight" sound similar.

4 → Bias

Higher error rate for female voices.

5 → Emotional Tone

Sarcasm missed by AI.

6 → Privacy

Private conversation recorded without consent.

Voice AI in the Real World



Voice Assistants

Siri, Alexa, Google Assistant: Answer questions, control smart homes, set reminders.



Audiobook Narration

TTS generates entire books in natural voices.



Accessibility Tools

Live captioning (STT) for deaf/hard of hearing; screen readers (TTS) for blind/low vision users.



Language Learning Apps

Evaluate pronunciation and provide feedback.



Customer Service Chatbots

Understand spoken questions and respond helpfully.



Real-time Translation Earbuds

Hear foreign language and speak translation in your ear.

Hands-On Activity: Create Your Own AI Voice with ElevenLabs

Goal: Use ElevenLabs to design a custom voice from a text description, then make it speak a sentence you choose.

Tool: ElevenLabs (free tier at elevenlabs.io)





Step 1: Open Voice Design

Log into ElevenLabs. Find **Voice Design** on the Home page or under the **Voices** section.



Step 2: Describe Your Voice

In the prompt box, type a short description of the voice you want to create. Click **Generate** and listen.



Step 3: Use in Text to Speech

Go to **Text to Speech**. Select your voice, type a sentence, click **Generate** and listen.



Step 4: Reflection

Answer reflection questions in your head or discuss with your mentor.

Hands-On Activity: Example Voice Prompts

In the prompt box, type a short description of the voice you want to create. **Example prompts (choose one or create your own):**

“

"A friendly wizard with a warm, slightly crackly voice."

”

“

"An energetic sports commentator who talks fast."

”

“

"A calm, soft voice for reading bedtime stories."

”

Click **Generate** and listen to the result.

Hands-On Activity:

Text to Speech

Go to **Text to Speech**. Select the voice you just created from the voice dropdown. Type a short sentence into the text box. For example:

- *"Hello! This is my custom AI voice."*
- *"I created this voice just by describing it."*

Click **Generate** and listen.

Reflection Questions

Answer these in your head or discuss with your mentor:

1. How close was the generated voice to what you imagined?
2. What do you think the AI did to turn your description into a voice?
3. Where might a tool like this be useful? Where could it be misused?

The Most Important Idea



ASR/STT

Sound Waves → Acoustic Features
→ Phonemes → Words



NLU

Figures out meaning and intent



TTS

Words → Tokens → Prosody →
Waveform

You are the human in the loop.

❏ **Voice AI is pattern matching, not true understanding, not true hearing.** Be aware of its limits and use it responsibly.

Summary:

Key Term	What It Means
Voice AI	AI that hears, interprets, and generates spoken language.
ASR/STT	Automatic Speech Recognition / Speech-to-Text – converts speech to text.
TTS	Text-to-Speech – converts text to speech.
Waveform	Visual graph of sound over time.

Summary:

Key Term	What It Means
Acoustic Features	Clues extracted from sound: pitch, energy, frequency patterns.
Phoneme	Smallest unit of sound in a language.
NLU	Understanding intent and meaning from text.
Prosody	The rhythm, pitch, and stress that make speech sound human.
Tokens	Small pieces words are broken into for processing.
Hear → Process → Speak	Voice AI framework.



Homework: Creative Project - From Idea to Audio

Use ChatGPT (or Google Gemini) and ElevenLabs together to create a short narrated story.

1. Brainstorm Ideas

Open ChatGPT or Gemini. Ask for 3 short story ideas using a prompt like: *"Give me 3 short story ideas about a mysterious sound heard at night. Each idea should be one sentence."* Pick your favorite.

2. Write the Story

Ask the AI to write a short story: *"Write a short story based on this idea: [paste idea]. Make it about 100–150 words. Use simple, vivid language."* Read the story. Edit anything you want - it's your story now.

3. Create a Voice

Log into ElevenLabs. Use **Voice Design** to create a voice that fits your story's mood. Example descriptions: *"A calm, gentle voice for a bedtime story."* / *"A mysterious, whispery voice."* / *"An excited, curious voice."*

4. Narrate Your Story

Go to **Text to Speech**. Select your custom voice. Paste your story and generate the audio. Listen to your story narrated by your own AI voice.

Homework: Reflection Questions

1

AI Collaboration

How did the AI help you with the story? What did you change or add?

2

Voice & Mood

Did the voice match the mood of your story? Why or why not?

3

Multi-Tool Learning

What's one thing you learned about working with multiple AI tools together?

Add to Your Prompt Library

Your New Prompt: "Explain how a voice assistant understands the command 'Set an alarm for 7 AM.' Use the Hear → Process → Speak framework. Describe the ASR/STT steps (Sound Waves → Acoustic Features → Phonemes → Words) and mention prosody in TTS."

Why This Prompt Works

Specific Example

It asks for a concrete command:
"Set an alarm for 7 AM."

Requires the Framework

Forces use of the Hear → Process
→ Speak structure.

Includes Sub-Steps

Includes the detailed sub-steps
you learned today.

Next Session Preview

Session 17: AI That Combines Everything - Multimodal AI

Now that you've seen how AI reads (NLP), sees (CV), and hears/speaks (Voice AI), next we'll explore **multimodal AI** - systems that understand text, images, audio, and video together.

